

The transshipment problem

The shortest path problem

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Optimization: principles and algorithms



The shortest path problem

The transshipment problems embeds a variety of other problems.

These problems inherit the properties of the transshipment problem.

In particular, the integrality of the optimal solution if the data is integer.

In this video, we introduce the shortest path problem as an instance of the transshipment problem.

Shortest path

Consider a network with nodes and arcs

Each arc is associated with a cost

Consider an origin and a destination

Find a simple forward path from o to d with the smallest cost

Shortest path



Shortest paths problems

In this video, single origin – single destination.

Shortest paths problems

- ▶ Single origin – single destination;
- ▶ Single origin – all destinations;
- ▶ All origins – single destination;
- ▶ All origins – all destinations.

Cost of a path

Definition

$$C(P) = \sum_{(i,j) \in P^{\rightarrow}} c_{ij} x_{ij} - \sum_{(i,j) \in P^{\leftarrow}} c_{ij} x_{ij}.$$

Forward path

$$C(P) = \sum_{(i,j) \in P^{\rightarrow}} c_{ij} x_{ij} = \sum_{(i,j) \in P} c_{ij} x_{ij}$$

Transshipment

Flow

Send one unit of flow from o to d .

Supply at o : $s_o = 1$

Supply at d : $s_d = -1$

Supply at any other node: $s_i = 0$

Total unimodularity: if there is a solution, flow on (i,j) : 0 or 1

Cost of a path

$$C(P) = \sum_{(i,j) \in P} c_{ij} x_{ij} = \sum_{(i,j) \in P} c_{ij}$$

Transshipment

- ▶ cost on each arc: c_{ij}
- ▶ supply at the origin: $s_o = 1$
- ▶ supply at the destination: $s_d = -1$
- ▶ supply at any other node: 0
- ▶ lower bound on each arc: 0
- ▶ upper bound on each arc: ∞

Transshipment

$$\min_{x \in \mathbb{R}^n} \sum_{(i,j) \in \mathcal{A}} c_{ij} x_{ij}$$

subject to

$$\sum_{j|(o,j) \in \mathcal{A}} x_{oj} - \sum_{k|(k,o) \in \mathcal{A}} x_{ko} = 1,$$

$$\sum_{j|(d,j) \in \mathcal{A}} x_{dj} - \sum_{k|(k,d) \in \mathcal{A}} x_{kd} = -1,$$

$$\sum_{j|(i,j) \in \mathcal{A}} x_{ij} - \sum_{k|(k,i) \in \mathcal{A}} x_{ki} = 0, \quad \forall i \in \mathcal{N}, i \neq o, i \neq d,$$

$$x_{ij} \geq 0, \quad \forall (i,j) \in \mathcal{A}.$$

Transshipment

Summary

The shortest path problem is a transshipment problem

It is equivalent to sending one unit of flow from the origin to the destination

The cost of a path is simply the sum of the cost of each arc along the path.